The evidence behind StrideEco

Published findings and local context support the concept; resident demand and repeat use still require a pilot.

Evidence	What it supports	Boundary of the claim
Physical activity is a national issue. NHMS 2023 reported 29.9% of Malaysian adults as physically inactive.[1]	A reason to explore practical ways to encourage regular walking and running.	National survey data; it does not establish inactivity rates among SDP residents or explain their causes.
Shade can influence route choice. A Singapore study found that sun/shade exposure affected pedestrian path choices in a hot urban environment.[2]	Offering a choice between route distance and estimated shade has a research basis.	No measured Elmina temperatures, shade accuracy or app retention can be inferred from this study.
Incentives can encourage activity. The BE ACTIVE randomised trial found increased walking with gamification and financial incentives.[3]	A rationale for testing a clear activity-to-reward loop.	Participants had cardiovascular disease or elevated risk. Drinking-water rewards for SDP residents were not tested.
Elmina has an active-living setting. SDP advertises a 300-acre Central Park and 90 km of cycling/jogging tracks in its City of Elmina material.[4]	A relevant township context for a bounded walking/running pilot.	Developer/masterplan information; present usable coverage, route access and visitor numbers have not been audited.
Organised running has taken place locally. A JomRun listing places an SDP 5 km / 10 km night run at Elmina Rainforest Knowledge Centre on 11 November 2023.[5]	Historical evidence of an organised running event in the township.	A historical event listing; it does not establish attendance, annual recurrence or demand for StrideEco.

Working problem statement: StrideEco will test whether route planning and a practical drinking-water reward help residents return to walking/running. Its planning insights would come from consent-based pilot activity data.

Superscript numbers refer to the source register on page 4.

Existing solutions & proposed contribution

The public features below establish precedents. This is a focused comparison, not an exhaustive competitor audit.

Existing approach	Publicly documented capability	Implication for StrideEco
Strava Sponsored Challenges[6]	Brands can reward completion of activity goals such as distance or time; reward claims lead to the partner brand's website.	Activity-based rewards are established. Points and rewards alone should not be presented as a new invention.
HPB Healthy 365 / Healthpoints[7]	Eligible programme activities, including steps and staying active, can earn Healthpoints redeemable for rewards.	The activity-to-points-to-reward pattern has an operational precedent. Feature availability does not prove StrideEco's effectiveness.
CoolWalks research[8]	Models shaded walking routes using building footprints and street networks; results depend on urban geometry and building heights.	Shade-aware routing has a research precedent. Local input quality and clear estimate labels are essential.

Proposed township-specific integration

PACE: Plan · Act · Collect · Exchange.
Plan a walking/running route → record eligible distance → collect points → redeem drinking water through a compatible existing machine operator.

Value to test: a useful reason to return, a traceable water-redemption process and consent-based evidence about frequently used pedestrian routes. No world-first, adoption or impact claim is made at this stage.

Pilot dependencies: confirmed operator, compatible redemption method, water stock, funding and an agreed points-to-water rule. Planning insights are a proposed output; live redemption and resident heatmaps are not established yet.

Public data & validation status

SUPPORTED = evidence available · PARTIAL = usable with stated limits · OPEN = pilot or partner confirmation required

Item / status	Evidence available now	What remains to establish
Map coverage PARTIAL	The team's 22 October 2026 OSM snapshot contains 88 eligible walking ways and 582 building footprints in a small Elmina park area.[9,10]	Current access, connectivity and path condition. This bounded extract does not cover all of City of Elmina.
Shade estimation PARTIAL	That snapshot contains zero building-height tags and zero tree-canopy features. Optional building shadows use an explicit assumed height.[10]	Defensible height/canopy inputs. Scenario results are not measured shade, temperature or a thermal-safety assessment.
Activity/reward precedent SUPPORTED	Research supports testing incentives; Strava and Healthy 365 document existing activity-reward systems.[3,6,7]	Whether distance-based points redeemable for water create repeat use among the intended residents.
Target-user demand OPEN	The resident persona and needs are team hypotheses. No resident interview findings, local Strava user count or retention baseline are verified.	Voluntary pilot feedback, repeat activity and reasons for continued use or drop-off.
Water redemption OPEN	The prototype can demonstrate the wallet and redemption journey. No suitable local machine/operator or live dispensing is confirmed.	Operator agreement, site, stock, funding, integration and successful end-to-end redemption checks.
Planning insights OPEN	Consent-based aggregation is proposed. Prototype map features are not resident usage observations.	Real consented activity, sufficient aggregated coverage and SDP review before interpreting infrastructure gaps.

Evidence rule: label published findings, prototype scenarios and team assumptions separately. Pilot targets and KPI definitions belong in A3; cost and funding assumptions belong in A4.

Source register

Public sources checked on 23 October 2026. Research results are interpreted within their original population, method and setting.

1. Institute for Public Health, Ministry of Health Malaysia
NHMS 2023: fact sheet.
Supports: National physical-inactivity statistic.
Official fact sheet

2. Melnikov et al. · Scientific Reports 12, 2441 (2022)
Behavioural thermal regulation explains pedestrian path choices in hot urban environments.
Supports: Singapore pedestrian route-choice evidence.
DOI / publisher

3. Fanaroff et al. · Circulation (2024)
BE ACTIVE randomised controlled trial: gamification and financial incentives.
Supports: Walking incentives in adults with cardiovascular disease or elevated risk.
Authors' institution summary · DOI

4. Sime Darby Property · City of Elmina
Official township webpage; undated.
Supports: Developer claims about the park and jogging/cycling network.
Official township page

5. JomRun · SDP Night Run & Ride
5 km / 10 km Neon Night Run; listed date 11 November 2023.
Supports: Historical Elmina event location and distance categories.
Historical event listing

6. Strava for Business
Sponsored Challenges; official product page.
Supports: Activity goals, incentives and partner-site reward claims.
Official feature page

7. HealthHub / Health Promotion Board, Singapore
Healthpoints and Healthy 365 rewards; official programme page.
Supports: Eligible activities earning points redeemable for rewards.
Official rewards page

8. Wolf, Vierø & Szell · Scientific Reports 15, 14911 (2025)
CoolWalks for active mobility in urban street networks.
Supports: Building-shadow routing method and geometric dependencies.
Primary manuscript · DOI

9. © OpenStreetMap contributors · ODbL 1.0
Bounded Elmina extract retrieved 22 October 2026.
Supports: Public map geometry; it is not a field inspection.
Raw map extract · Licence / attribution

10. StrideEco team · prototype audit (22 October 2026)
public/data/map-provenance.json and docs/data-sources.md.
Supports: Local extract counts, missing tags and scenario limitations. Team-produced technical evidence; not independent research.